$$= \hat{i}\left(A_y B_z - A_z B_y\right) + \hat{j}\left(A_z B_x - A_x B_z\right) + \hat{k}\left(A_x B_y - A_y B_x\right)$$

Note that in the $\hat{j}^{th}$ component the order of multiplication is different than $\hat{i}^{th}$ and $\hat{k}^{th}$ components.

(ix) If two vectors $\vec{A}$ *and* $\vec{B}$ form adjacent sides in a parallelogram, then the magnitude of $\vec{A} \times \vec{B}$ will give the area of the parallelogram as represented graphically in Figure 2.23.

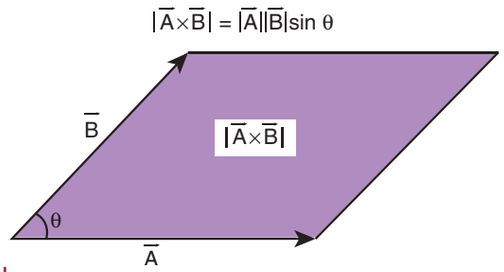


Figure 2.23 Area of parallelogram

(x) Since we can divide a parallelogram into two equal triangles as shown in the Figure 2.24, the area of a triangle with $\vec{A}$ *and* $\vec{B}$ as sides is $\frac{1}{2}\left|\vec{A} \times \vec{B}\right|$. This is shown in the Figure 2.24. (This fact will be used when we study Kepler's laws in unit 6)

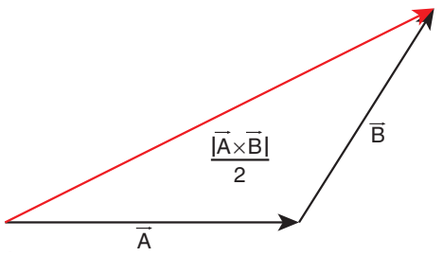


Figure 2.24 Area of triangle

A number of quantities used in Physics are defined through vector products. Particularly physical quantities representing rotational effects like torque, angular momentum, are defined through vector products.

Examples

(i) Torque $\vec{\tau} = \vec{r} \times \vec{F}$. where $\vec{F}$ is Force and $\vec{r}$ is position vector of a particle

(ii) Angular momentum $\vec{L} = \vec{r} \times \vec{p}$ where $\vec{p}$ is the linear momentum

(iii) Linear Velocity $\vec{v} = \vec{\omega} \times \vec{r}$ where $\vec{\omega}$ is angular velocity

EXAMPLE 2.9

Two vectors are given as $\vec{r} = 2\hat{i} + 3\hat{j} + 5\hat{k}$ and $\vec{F} = 3\hat{i} - 2\hat{j} + 4\hat{k}$. Find the resultant vector $\vec{\tau} = \vec{r} \times \vec{F}$

Solution

$$\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 5 \\ 3 & -2 & 4 \end{vmatrix}$$

$$\vec{\tau} = (12 - (-10))\hat{i} + (15 - 8)\hat{j} + (-4 - 9)\hat{k}$$

$$\vec{\tau} = 22\hat{i} + 7\hat{j} - 13\hat{k}$$

2.5.3 Properties of the components of vectors

If two vectors $\vec{A}$ and $\vec{B}$ are equal, then their individual components are also equal.

Let $\vec{A} = \vec{B}$

Then $A_x\hat{i} + A_y\hat{j} + A_z\hat{k} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$

i.e., $A_x = B_x$, $A_y = B_y$, $A_z = B_z$

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